Tema 5

a) Pt $t=1$: $\alpha_1(j) = P(O_1 = o_1, X_1 = j)$

$\alpha_1(j) = P(X_1 = j) \cdot P(O_1 = o_1 / X_1 = j)$ |

$P_{(X_1 = j)} = \tilde{\pi}_j$ |

$P_{(O_1 = o_1 | X_1 = j)} = b_j(o_1)$ |

$\Rightarrow \alpha_1(j) = \tilde{\pi}_j \cdot b_j(o_1)$

b) $\alpha_t(j) = P(O_1 = o_1, \ldots, O_t = o_t, X_t = j, X_{t-1} = i \Rightarrow$

$\Rightarrow \alpha_t(j) = \sum_i P(O_1 = o_1, \ldots, O_t = o_t, X_{i-1} = i, X_t = j)$

$\alpha_t(j) = \sum_i P(O_t = o_t / X_t = j) \cdot P(X_t = j / X_{t-1} = i) \cdot$

$P(O_1 = o_1, \ldots, O_{t-1} = o_{t-1}, X_{t-1} = i)$

$P(O_t = o_t / X_t = j) = b_j(o_t)$ |

$P(X_t = j / X_{t-1} = i) = a_{ij}$ |

$\Rightarrow \alpha_t(j) = \sum_i b_j(o_t) \cdot a_{ij} \cdot \alpha_{t-1}(i)$

$\alpha_t(j) = b_j(o_t) \cdot \sum_i \alpha_{t-1}(i) \cdot a_{ij}$, pt. $t = 2, \ldots, \overline{T}$

c) $P(O_1 = o_1, \dots, O_t = o_i) = \sum_j P(O_1 = o_1, \dots, O_t = o_i, X_t = j)$

$P(O_1 = o_1, \dots, O_t = o_{t_3}, X_t = j) = \alpha_t(j)$ ⇒

⇒ $P(O_1 = o_1, \dots, O_t = o_t) = \sum_j \alpha_t(j)$